ClimaGuard AI – QA Test Cases

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Overview

This checklist holds 65 test cases that cover every acceptance criterion of the 15 user stories in the backlog (ClimaGuard AI – Epics and User Stories), plus the PRD's non-functional requirements and model metrics. QA sets each test's Status as it runs; a story passes only when all of its tests pass.

Test types: *Positive* checks the expected path. *Negative* checks that wrong input or access is refused. *Boundary* checks the edges of a rule (thresholds, limits, counts).

Test environment

- Demo laptop with the app running from the README, using cached CSV and GeoJSON files.
- Browser: Chrome, at 1366 × 768 for desktop tests and 360 px wide for phone tests.
- A stopwatch (phone timer) for the 3-second and 5-minute checks.

Demo accounts

Account	Role	Scope
admin@climaguard.test	State admin	All districts
kavitha@climaguard.test	Deputy Director of Health Services	Krishnagiri
Selvi, mobile 90000 00001	Village Health Nurse	Shoolagiri block, Krishnagiri
Ravi, mobile 90000 00002	Village Health Nurse	Hosur block, Krishnagiri

Accounts are seeded test data with made-up contact details; they are not real people.

Data fixtures

Dev prepares these three cached runs so the alert and map rules can be tested on demand.

Fixture	What it contains	Used by
F1 – Alert scenario	Krishnagiri High on run 1 and run 2; Madurai High on run 2 only; Coimbatore jumps Low → High in one run; The Nilgiris Low	E2, E3, E4
F2 – Quiet week	No district at High or above	Empty-state tests
F3 – Threshold probe	Five districts with probabilities 0.24, 0.25, 0.50, 0.75 and 0.76	Risk-level boundary test

Summary and exit criteria

Run the Must-Have tests first (E1-S4, E2, E3, E4-S1 to S3); they are the ones the demo depends on.

Area	Stories covered	Test cases	Priority
E1 Platform, data and access	E1-S1 to S4	16	Should (S1–S3), Must (S4)
E2 Risk outlook map	E2-S1 to S3	8	Must
E3 District explanation	E3-S1, S2	6	Must
E4 Field priority and alerts	E4-S1 to S4	17	Must (S1–S3), Could (S4)
E5 Planning aids	E5-S1, S2	7	Should
Non-functional and model metrics	PRD NF1–NF6, model and product metrics	11	Must
Total	**15 stories**	**65**	

Ready to demo when:

- [x] ~~Every Must-Have test case is Pass.~~
- [x] ~~No open defect of Critical severity.~~
- [x] ~~The full demo path passes with Wi-Fi off (TC-NF-01).~~
- [x] ~~Any Fail on a Should or Could test is logged in the defect log and the feature is left out of the demo.~~

E1 – Platform, data and access

Account and sign-in tests (E1-S1 to S3) prove the start of the flow; the pipeline tests (E1-S4) prove every risk level the demo shows.

ID	Story	Type	Scenario and steps	Expected result	Status
TC-E1-01	E1-S1	Positive	**Admin creates an officer.** Sign in as admin → User management → Add officer: name, kavitha@climaguard.test, Krishnagiri → Save	Account listed with role Deputy Director and district Krishnagiri	Not run
TC-E1-02	E1-S1	Negative	**Duplicate email.** Repeat TC-E1-01 with the same email	Message "An account with this email already exists"; user count unchanged	Not run
TC-E1-03	E1-S1	Positive	**Temporary password shown once.** Create an account → note the password → refresh the page → open the users data file	Password shown once only; the data file holds a hash, not the plain password	Not run
TC-E1-04	E1-S1	Negative	**Missing fields.** Add officer with the district left empty → Save	Save blocked with a field error; no account created	Not run
TC-E1-05	E1-S2	Positive	**Block picker scope.** Sign in as Kavitha → Add field staff → open the block picker	Only Krishnagiri's blocks are listed	Not run
TC-E1-06	E1-S2	Positive	**Create nurse ID.** Enter name Selvi, mobile 90000 00001, block Shoolagiri → Save	Staff ID created and listed with Shoolagiri	Not run
TC-E1-07	E1-S2	Negative	**Duplicate mobile.** Add another nurse with mobile 90000 00001 → Save	Error shown; no new staff ID	Not run
TC-E1-08	E1-S2	Boundary	**No block chosen.** Add a nurse with zero blocks selected → Save	Save blocked: at least one block is required	Not run
TC-E1-09	E1-S3	Positive	**Role landing.** Sign in as admin, then Kavitha, then Selvi	Admin → user management; Kavitha → risk map; Selvi → "My blocks"	Not run
TC-E1-10	E1-S3	Positive	**First sign-in.** Sign in as a new user with the temporary password	Must set a new password before any other screen opens	Not run
TC-E1-11	E1-S3	Negative	**Wrong password.** Enter a valid email with a wrong password	Error message; no data shown	Not run
TC-E1-12	E1-S3	Negative	**Deep link while signed out.** Sign out → paste the risk map URL	Redirected to sign-in	Not run
TC-E1-13	E1-S3	Negative	**Wrong role by URL.** Sign in as Selvi → open the user management URL	Access refused; nothing from the admin screen shown	Not run

ID	Story	Type	Scenario and steps	Expected result	Status
TC-E1-14	E1-S4	Positive	**Pipeline runs offline.** Turn Wi-Fi off → run the pipeline → open its output	All 38 districts have a probability and a level for 7 and 14 days	Not run
TC-E1-15	E1-S4	Boundary	**Risk thresholds.** Run fixture F3 (0.24, 0.25, 0.50, 0.75, 0.76)	Low, Moderate, High, High, Very high (lower bound inclusive; confirm with Dev, see Q1)	Not run
TC-E1-16	E1-S4	Positive	**Backtest report.** Run the backtest on the held-out season	Report shows detection rate, lead time and Brier score next to the "same month last year" baseline	Not run

E2 – Risk outlook map

Run with fixture F1 loaded unless a test says otherwise.

ID	Story	Type	Scenario and steps	Expected result	Status
TC-E2-01	E2-S1	Positive	**Map shows every district.** Sign in as Kavitha → open the risk map → count districts	38 districts, each with a colour and a text label (Low, Moderate, High, Very high)	Not run
TC-E2-02	E2-S1	Positive	**Disclaimer visible.** Open the map at 1366 × 768 without scrolling	"Risk estimate, not a diagnosis" is visible	Not run
TC-E2-03	E2-S1	Positive	**Open a district.** Click Krishnagiri on the map	Krishnagiri's detail view opens (E3-S1)	Not run
TC-E2-04	E2-S2	Positive	**Switch outlook in time.** On the 7-day view, choose 14 days and start the stopwatch	Every district updates within 3 seconds	Not run
TC-E2-05	E2-S2	Positive	**Outlook named.** View the 7-day and then the 14-day map	The screen title states which outlook is shown	Not run
TC-E2-06	E2-S3	Positive	**Needs-attention list.** Open the list with F1 loaded	Only High and Very high districts, sorted by probability, highest first	Not run
TC-E2-07	E2-S3	Positive	**List matches map.** Compare the list with the map for the five spotlight districts	Same districts and same levels in both	Not run
TC-E2-08	E2-S3	Boundary	**Quiet week.** Load fixture F2 → open the list	Message "No districts need attention this week"	Not run

E3 – District explanation

ID	Story	Type	Scenario and steps	Expected result	Status
TC-E3-01	E3-S1	Positive	**Three drivers.** Open Krishnagiri's detail view	Exactly three drivers, ordered by contribution, largest first	Not run
TC-E3-02	E3-S1	Positive	**Level matches map.** Compare the detail view's level and probability with the map	Same level and probability in both places	Not run
TC-E3-03	E3-S1	Positive	**Drivers follow the data.** Note the drivers → swap in a drier rainfall file → rerun the pipeline → reopen	Drivers and their order change to match the new data	Not run
TC-E3-04	E3-S2	Positive	**Card content.** Read each of the three cards	Each names the weather factor, whether it raises or lowers risk, and the time lag	Not run
TC-E3-05	E3-S2	Negative	**No raw names.** Open the detail view of all 38 districts	No card shows a code-style name such as rain_lag_3w	Not run
TC-E3-06	E3-S2	Boundary	**Lookup covers the model.** Compare the model's feature list with the lookup table	Every feature has a plain-language sentence; none missing	Not run

E4 – Field priority and alerts

The alert-rule tests (TC-E4-05 to 08) need fixture F1, which holds two pipeline runs.

ID	Story	Type	Scenario and steps	Expected result	Status
TC-E4-01	E4-S1	Positive	**Priority list for a High district.** As Kavitha, open Krishnagiri → Priority list	Blocks ranked by district risk × vulnerability index; every block has at least one action	Not run
TC-E4-02	E4-S1	Positive	**Ranking is correct.** Take the top 3 blocks' scores and recompute risk × vulnerability in a spreadsheet	Order on screen matches the spreadsheet	Not run
TC-E4-03	E4-S1	Negative	**No list below High.** Open The Nilgiris (Low)	No priority list is shown	Not run
TC-E4-04	E4-S1	Positive	**Rank is explained.** Open any block row	Density, share of children and older adults, and past hotspots are shown	Not run
TC-E4-05	E4-S2	Positive	**Two High runs fire an alert.** Run F1 run 1 then run 2 → open the outbox	One alert for each nurse assigned to a Krishnagiri priority block (Selvi; Ravi too if Hosur is listed)	Not run

ID	Story	Type	Scenario and steps	Expected result	Status
TC-E4-06	E4-S2	Positive	**Two-level jump fires an alert.** Check Coimbatore after F1 (Low → High in one run)	An alert is created	Not run
TC-E4-07	E4-S2	Negative	**One High run, no alert.** Check Madurai after F1 (High on run 2 only)	No alert is created	Not run
TC-E4-08	E4-S2	Boundary	**One-level step, no alert.** Set a district Moderate → High for one run only	No alert is created	Not run
TC-E4-09	E4-S2	Positive	**Message content.** Open Selvi's outbox message	Shows recipient and time; message is 160 characters or fewer and names the block and the first action	Not run
TC-E4-10	E4-S2	Negative	**No personal data.** Read every outbox message from F1	No patient names, phone numbers or case details in any message	Not run
TC-E4-11	E4-S3	Positive	**My blocks.** Sign in as Selvi	Only Shoolagiri (her block) shown, latest alert at the top	Not run
TC-E4-12	E4-S3	Positive	**Age-specific advice.** Open Shoolagiri in My blocks	What to check, plus separate advice for children (0–12), adults (13–64) and older adults (65+)	Not run
TC-E4-13	E4-S3	Boundary	**Nothing to visit.** Load fixture F2 → sign in as Selvi	Message "No priority visits this week"	Not run
TC-E4-14	E4-S3	Negative	**Another nurse's blocks.** As Selvi, open Ravi's Hosur block by changing the URL	Access refused; Hosur's details not shown	Not run
TC-E4-15	E4-S4	Positive	**Mark visited.** As Selvi, mark Shoolagiri visited with 0 breeding sites found	Shown as visited with today's date; 0 is accepted	Not run
TC-E4-16	E4-S4	Boundary	**Invalid site count.** Enter −1, then the text "abc", as sites found	Both refused with a clear message	Not run
TC-E4-17	E4-S4	Positive	**Officer sees coverage.** After TC-E4-15, sign in as Kavitha → open Krishnagiri	"X of Y priority blocks visited" and the sites found per block	Not run

E5 – Planning aids

These are Should-Have tests; run them only after every Must-Have test passes.

ID	Story	Type	Scenario and steps	Expected result	Status
TC-E5-01	E5-S1	Positive	**What-if result.** Open Krishnagiri → move rainfall to +30% → start the stopwatch	New level and drivers appear next to the current ones within 3 seconds	Not run
TC-E5-02	E5-S1	Boundary	**Slider limits.** Drag rainfall and humidity past both ends	Values stop at −50% and +50%	Not run
TC-E5-03	E5-S1	Positive	**Scenario label.** Change any slider	Screen shows "Scenario, not forecast"	Not run
TC-E5-04	E5-S1	Positive	**Reset.** After TC-E5-01, press Reset	Sliders return to forecast values and the original level returns	Not run
TC-E5-05	E5-S1	Negative	**Scenario stays private.** Run a what-if → open the map, the needs-attention list and the outbox	Map, list and alerts are unchanged; no alert from the scenario	Not run
TC-E5-06	E5-S2	Positive	**Predicted vs reported.** Open the panel	Each month of the held-out season shows the predicted probability next to whether it was above normal	Not run
TC-E5-07	E5-S2	Positive	**Panel matches backtest.** Compare the panel's figures with the TC-E1-16 report	Detection rate, lead time and season name match the report exactly	Not run

Non-functional and model metrics

These come from the PRD's non-functional requirements (NF1–NF6) and its model and product metrics.

ID	Requirement	Type	Scenario and steps	Expected result	Status
TC-NF-01	NF1 Offline	Positive	**Full demo offline.** Turn Wi-Fi off → run the demo path from sign-in to visit check-off	Every step works with no network error	Not run
TC-NF-02	NF2 No personal data	Negative	**Data review.** Open every data file and three API responses (/risk, /priority, /alerts)	Only aggregated counts; no patient names, addresses or case records	Not run
TC-NF-03	NF3 Disclaimer	Positive	**Every screen.** Visit map, district view, priority list, My blocks, outbox, what-if, feedback panel	"Risk estimate, not a diagnosis" on each	Not run

ID	Requirement	Type	Scenario and steps	Expected result	Status
TC-NF-04	NF4 Phone width	Positive	**360 px screen.** Resize Chrome to 360 px → open My blocks and the map	No sideways scrolling; text readable; buttons tappable	Not run
TC-NF-05	NF5 Not colour alone	Positive	**Grayscale check.** Take a grayscale screenshot of the map	Every district's level can still be read from its text label	Not run
TC-NF-06	NF6 Traceability	Positive	**Story IDs.** For each screen and endpoint, search the code and commits for its story ID	Every screen and endpoint maps to a story ID	Not run
TC-NF-07	Product metric	Boundary	**Map load time.** Clear the cache → open the map → stopwatch	Map loads in under 3 seconds	Not run
TC-NF-08	Product metric	Positive	**Stranger setup.** Someone outside the team follows the README on a fresh laptop	App running in under 5 minutes	Not run
TC-ML-01	Model metric	Positive	**Detection rate.** Read the backtest report (TC-E1-16)	Detection rate higher than "same month last year"	Not run
TC-ML-02	Model metric	Boundary	**Lead time.** Read the backtest report	2 weeks or more for at least half of the above-normal months	Not run
TC-ML-03	Model metric	Positive	**Probability accuracy.** Read the backtest report	Brier score lower than the baseline	Not run

If a model metric fails, record the real result in the defect log and present it honestly; it does not block the demo (PRD risk R3).

Defect log

Log every Fail here, then set the test's Status to Fail and link the defect ID in a comment on that row.

Severity: *Critical* breaks the demo path or shows wrong risk or personal data. *Major* breaks a story's acceptance criterion but has a workaround. *Minor* is cosmetic.

Defect ID	Test case	Severity	What happened	Steps to reproduce	Status
D-01		▼			▼
D-02		▼			▼
D-03		▼			▼

Open questions for Dev

- [ ] **Q1 – Threshold edges:** the PRD says Moderate 25–50% and High 50–75%. Is exactly 50% High and exactly 75% High? TC-E1-15 assumes the lower bound is inclusive.
- [ ] **Q2 – Alert recipients:** does an alert go to every nurse in the district, or only to nurses whose blocks are on the priority list? TC-E4-05 assumes the second.
- [ ] **Q3 – Fixtures:** can Dev ship fixtures F1–F3 as files the app can switch between, so QA can rerun the alert and threshold tests?